BME lecture week:3

Faculty of Engineering

Department of Electrical and Electronic Engineering

(AIUB)

Course convenors: Dr. Mehedi Bin Mohammad, Safayet Ahmed and

This class tutor: Mafruha Farzana

Room: F4-13, Level 4,

Campus- 2,

Extension: 265,

Email: mfarzana20@aiub.edu,

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Textbooks:

[1] R.S. Khurmi & J.K.Gupta, “A Text Book of Thermal Engineering”, two colour edition 1978(Reprint 2003), S.Chand & Company Ltd.

[2] R.S. Khurmi, “A Text Book of Hydraulics, Fluid Mechanics and Hydraulic Machines”, S.Chand & Company Ltd, Reprint 2007

References:

[1] V.P.Vasandani & D.S. Kumar, “Heat Engineering”, Metropolitan Book Co.(Pvt.) Ltd. Reprint 2001

[2] M.J. Moran & H.N. Shapiro, “Engineering Thermodynamics”, John Wiley and Sons.2000

Course contents of first 3 weeks

-Chapter 10 for week three,
-Chapters 6 and 26 for week four,

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Thermodynamic vapour cycle

We have already discussed the thermodynamic cycles of air in chapter 6. But in this chapter, we shall discuss the thermodynamic cycles of vapour. In a vapour cycle, all the theory remains the same, except the working substance, which is steam. The steam may be in any form *i.e.* wet, dry saturated or superheated. Though there are many vapour cycles, yet the following are important from the subject point of view :

10.2. Carnot Cycle with Steam as Working Substance

The schematic diagram of a Carnot engine is shown in Fig. 10.1 and the Carnot cycle using steam as the working substance is represented on p-v and T-s diagrams as shown in Fig. 10.2 (a) and (b) respectively.

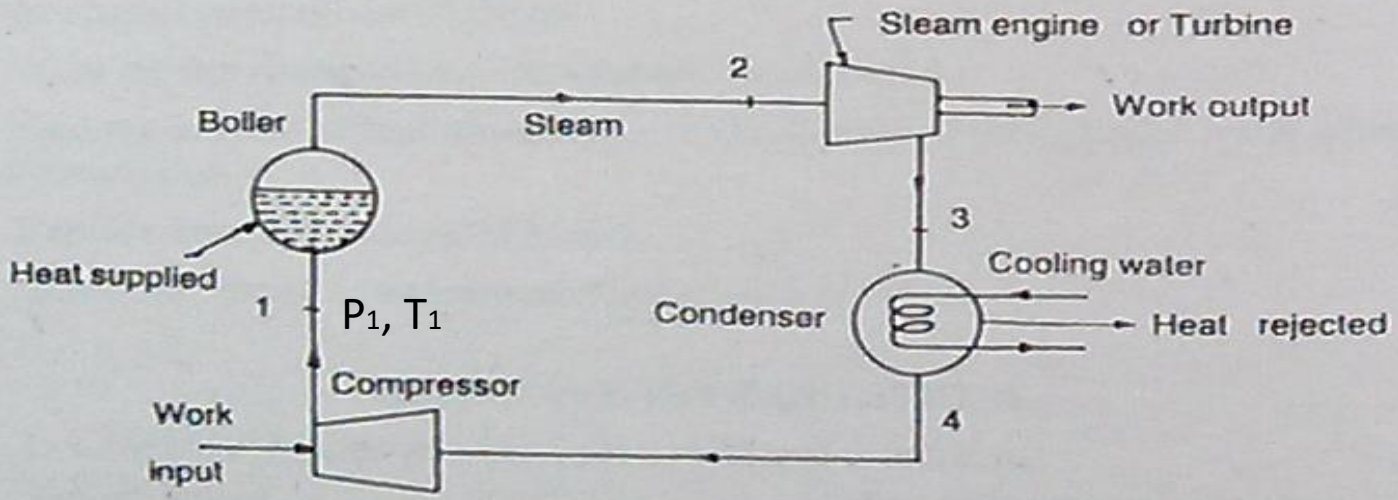


Fig. 10.1. Schematic diagram of a Carnot engine.

Consider 1 kg of saturated water at pressure p_1 and absolute temperature T_1, as represented by point 1 in Fig. 10.2 (a) and (b). The cycle is completed by the following four processes :

1. *Process 1-2* ...

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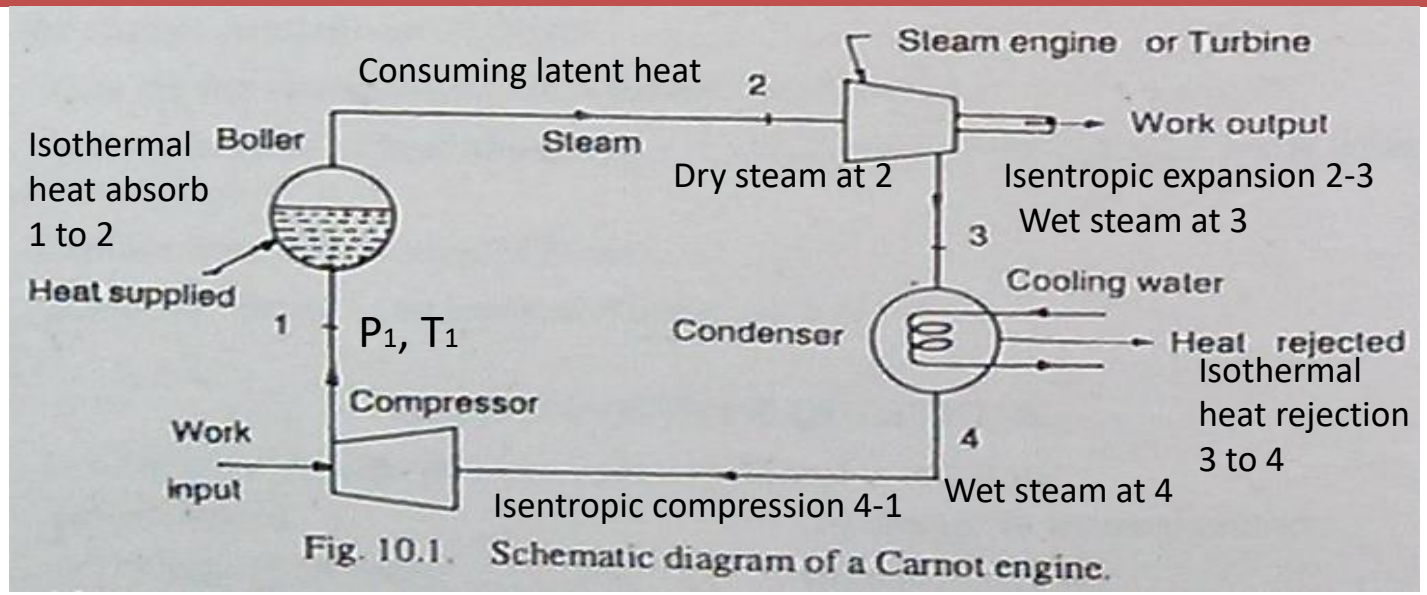
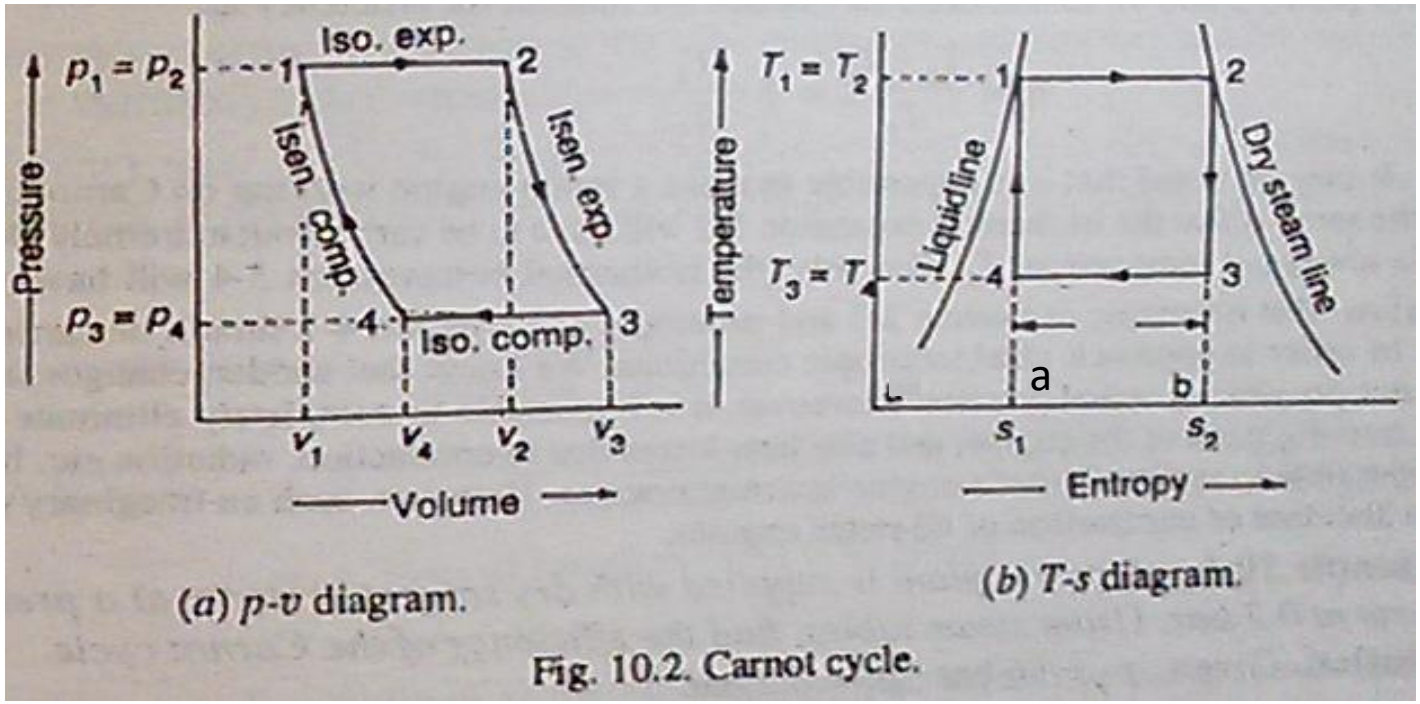


Fig. 10.1. Schematic diagram of a Carnot engine.



(a) p-v diagram. (b) T-s diagram.

Fig. 10.2. Carnot cycle.

A **Carnot heat engine** is an engine that operates on the reversible **Carnot cycle**. The Carnot cycle is a **theoretical thermodynamic cycle**. A thermodynamic cycle consists of a linked sequence of **thermodynamic processes** that involve transfer of heat and work into and out of the system, while varying P, T, and other state variables within the system, and that eventually returns the system to its initial state.

At **constant Temperature** the state changes.

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1. *Process 1-2.* The saturated water at point 1 is isothermally converted into dry saturated steam, in a boiler, and the heat is absorbed at a constant temperature T_1 and pressure p_1. The dry state of steam is represented by point 2. It means that the temperature T_2 (*i.e.* at point 2) and pressure p_2 (*i.e.* at point 2) is equal to temperature T_1 and pressure p_1 respectively. This isothermal process is represented by curve 1-2 on p-v and T-s diagrams in Fig. 10.2 (*a*) and (*b*).

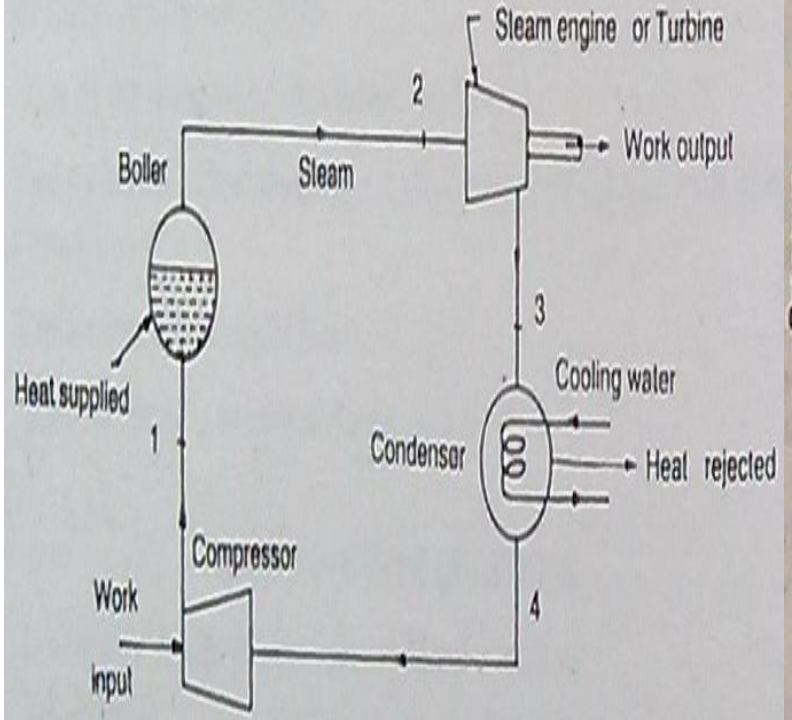


Fig. 10.1. Schematic diagram of a Carnot engine.

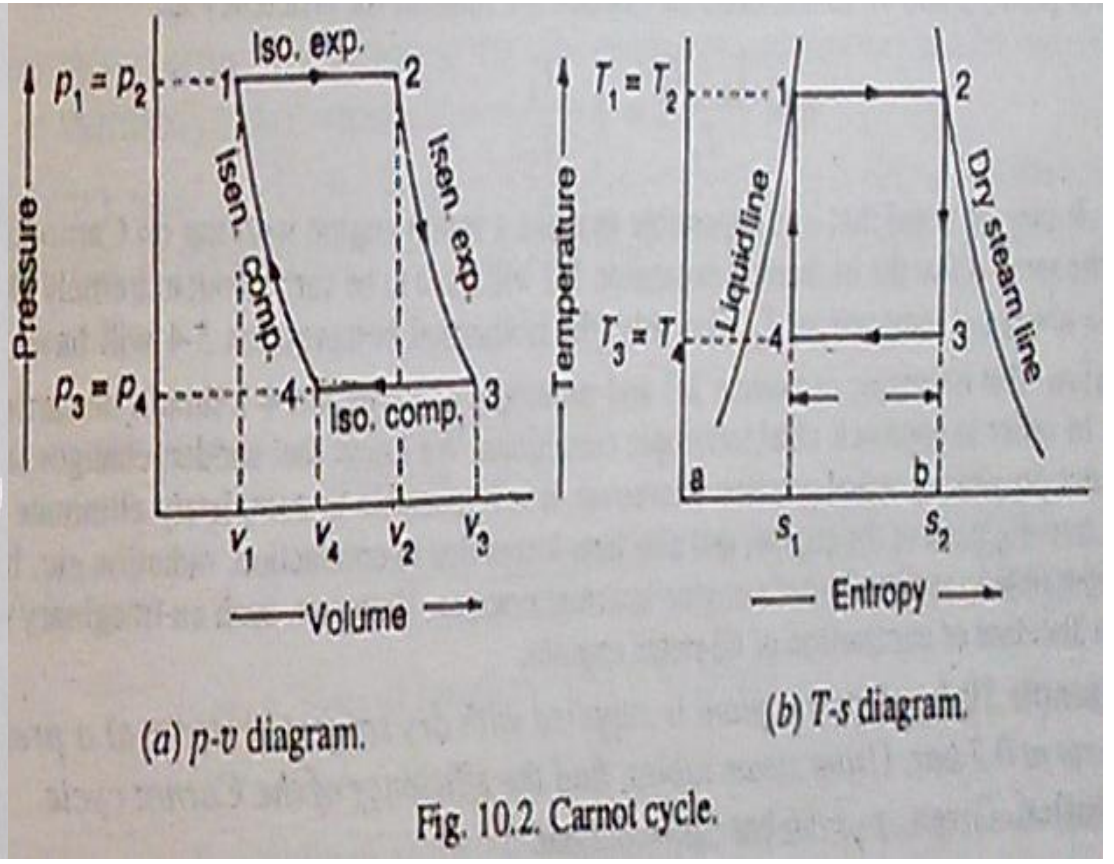


Fig. 10.2. Carnot cycle.

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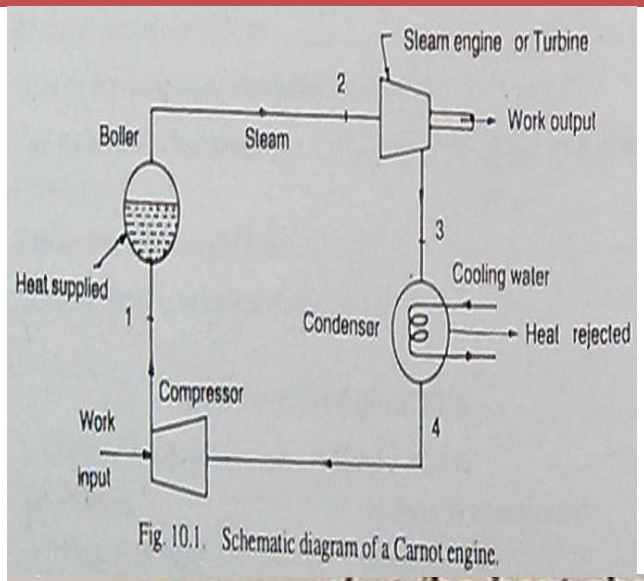


Fig. 10.1. Schematic diagram of a Carnot engine.

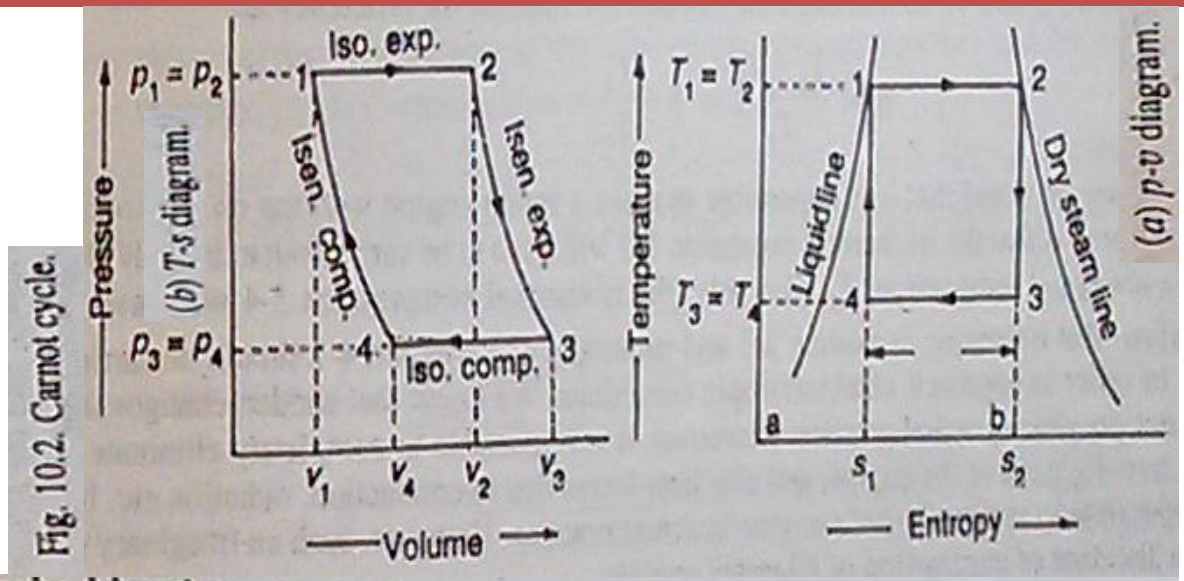


Fig. 10.2. Carnot cycle.

We know that the heat absorbed by the saturated water during its conversion into dry steam is its latent heat of evaporation (*i.e.* $h_{fg1} = h_{fg2}$) corresponding to a pressure p_1 or p_2 ($p_1 = p_2$).

We also know that the area 1-2-*b*-*a* in the *T*-*s* diagram represents the heat absorbed to some scale, during the isothermal process.

∴ Heat absorbed during isothermal process (area 1-2-*b*-*a*),

$$q_{1-2} = \text{Change in entropy} \times \text{Absolute temperature}$$

$$= (s_2 - s_1)\, T_1 = (s_2 - s_1)\, T_2 \qquad (\because\ T_1 = T_2) \quad \ldots (i)$$

2. *Process 2-3.* The dry steam at point 2 now expands isentropically in a steam engine or turbine. The pressure and temperature falls from p_2 to p_3 and T_2 to T_3 respectively. Since no heat is supplied or rejected during this process, therefore there is no change of entropy. The isentropic expansion is represented by the curve 2-3 as shown in Fig. 10.2 (*a*) and (*b*).

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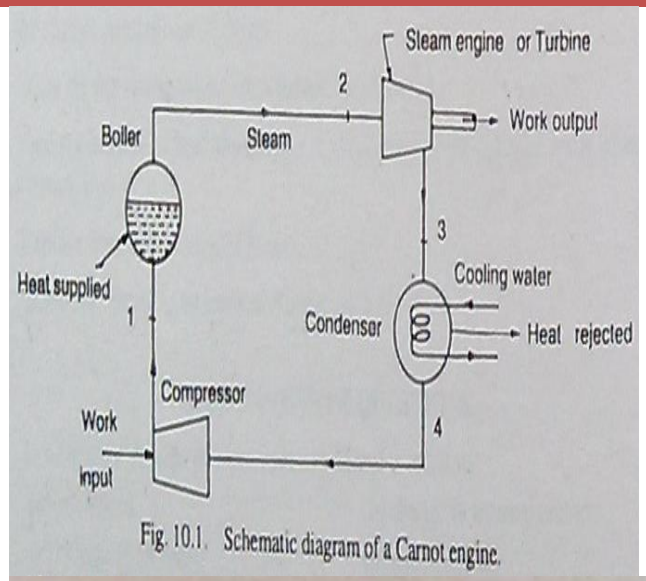


Fig. 10.1. Schematic diagram of a Carnot engine.

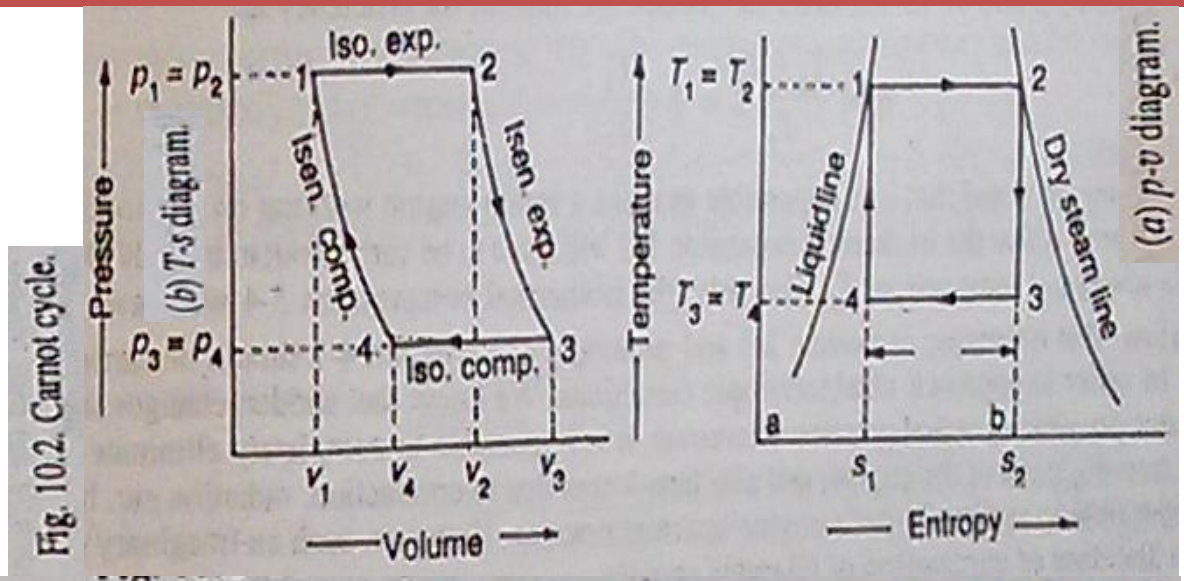


Fig. 10.2. Carnot cycle.

3. *Process 3-4.* The wet steam at point 3 is now isothermally condensed in a condenser and the heat is rejected at a constant temperature T_3 and pressure p_3. It means that the temperature T_4 (*i.e.* the heat is rejected at a constant temperature T_3 and pressure p_3. It means that the temperature T_4 (*i.e.* at point 4) and pressure p_4 (*i.e.* at point 4) is equal to the temperature T_3 and pressure p_3 respectively. This isothermal process is represented by the curve 3-4 on *p-v* and *T-s* diagrams as shown in Fig. 10.2 (*a*) and (*b*).

We know that area 3-4-*b*-*a* in the *T-s* diagram represents the heat rejected to some scale during the isothermal process.

∴ Heat rejected during isothermal compression (area 3-4-*b*-*a*),

$$q_{3-4} = (s_2 - s_1)\, T_3 = (s_2 - s_1)\, T_4 \quad \ldots (\because T_3 = T_4) \quad \ldots (ii)$$

...isentropically in a compressor,

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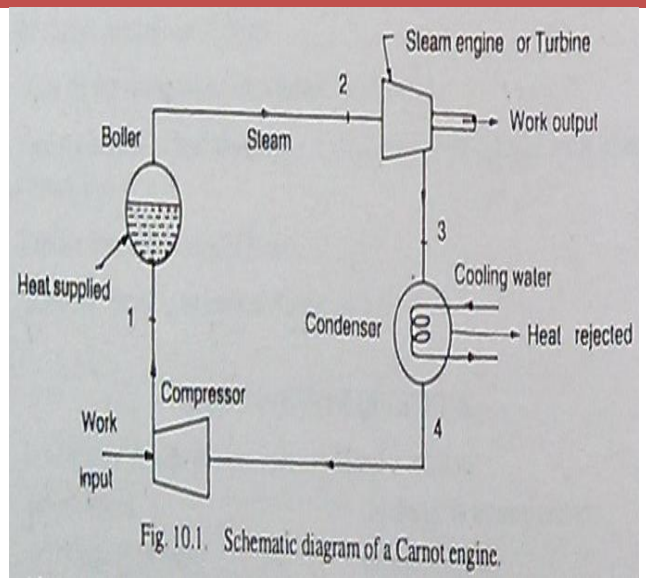


Fig. 10.1. Schematic diagram of a Carnot engine.

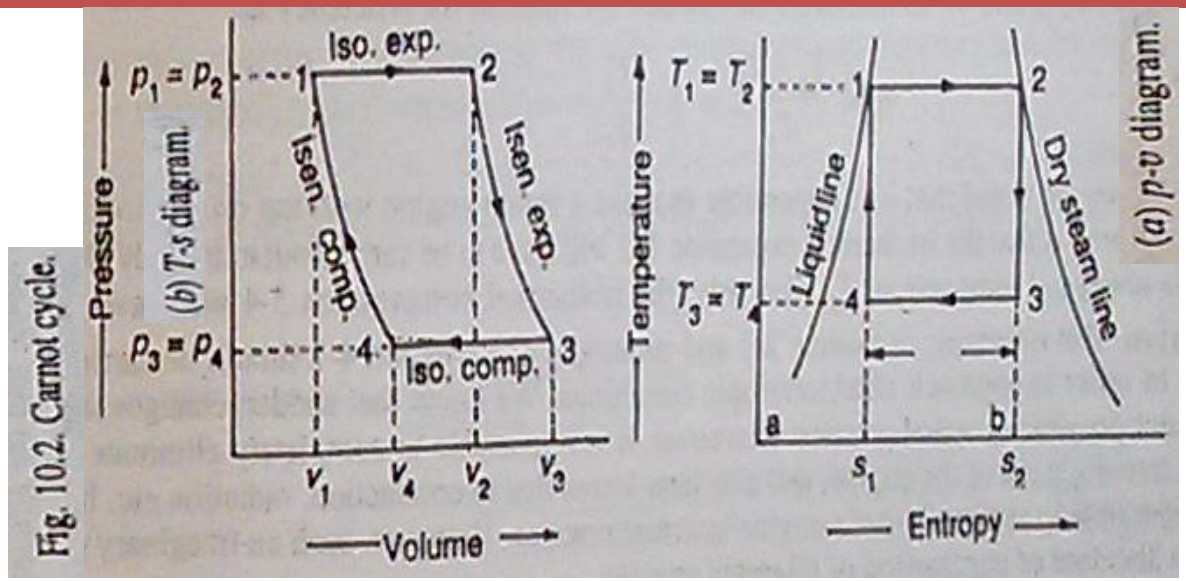


Fig. 10.2. Carnot cycle.

$q_{3-4} = (s_2 - s_1) T_3$

4. Process 4-1. The wet steam at point 4 is finally compressed isentropically in a compressor, till it returns back to its original state (point 1). The pressure and temperature rises from p_4 to p_1 and T_4 to T_1 respectively. The isentropic compression is represented by the curve 4-1 as shown in Fig. 10.2 (*a*) and (*b*). Since no heat is absorbed or rejected during this process, therefore entropy remains constant. This completes the cycle.

We know that work done during the cycle

= Heat absorbed – Heat rejected

$= (s_2 - s_1) T_1 - (s_2 - s_1) T_3 = (s_2 - s_1) (T_1 - T_3)$

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and efficiency of the Carnot cycle,

$$\eta = \frac{\text{Work done}}{\text{Heat absorbed}}$$

$$= \frac{(s_2 - s_1)(T_1 - T_3)}{(s_2 - s_1) T_1} = \frac{T_1 - T_3}{T_1} = 1 - \frac{T_3}{T_1}$$

where T_1 = Highest temperature corresponding to the boiler pressure $p_1 = p_2$, and

T_3 = Lowest temperature corresponding to the condenser pressure $p_3 = p_4$

Notes : 1. Since the heat absorbed is at the highest temperature and rejected at the lowest temperature, the Carnot cycle would give a maximum possible efficiency

2. In the above theory, we have taken temperature at points 1, 2, 3 and 4 as T_1, T_2, T_3 and T_4 respectively in order to keep similarity between Carnot cycle and other cycles. But some authors take it T_1 (for points 1 and 2) and T_2 (for points 3 and 4). In that case, they obtain the relation for efficiency as,

$$\eta = \frac{T_1 - T_2}{T_1} = 1 - \frac{T_2}{T_1}$$

3. It may be noted that it is impossible to make a steam engine working on Carnot cycle. The simple reason for the same is that the isothermal expansion 1-2 will have to be carried out extremely slow to ensure that the steam is always at temperature T_1. Similarly, the isothermal compression 3-4 will have to be carried out extremely slow. But isentropic expansion 2-3 and isentropic compression 4-1 should be carried out as quickly as possible in order to approach ideal isentropic conditions. We know that sudden changes in the speed of an engine are not possible in actual practice. Moreover, it is impossible to completely eliminate friction between the various moving parts of the engine, and also heat losses due to conduction, radiation etc. It is thus obvious that it is impossible to realise Carnot's engine in actual practice. However, such an imaginary engine is used as the ultimate standard of comparison of all steam engines.

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10.4. Rankine Cycle

The Rankine cycle is an ideal cycle for comparing the performance of steam plants. It s modified form of Carnot cycle, in which the condensation process (3-4) is continued until the steam is condensed into water. The schematic diagram of a steam engine or a turbine plant is shown in Fig. 10.3.

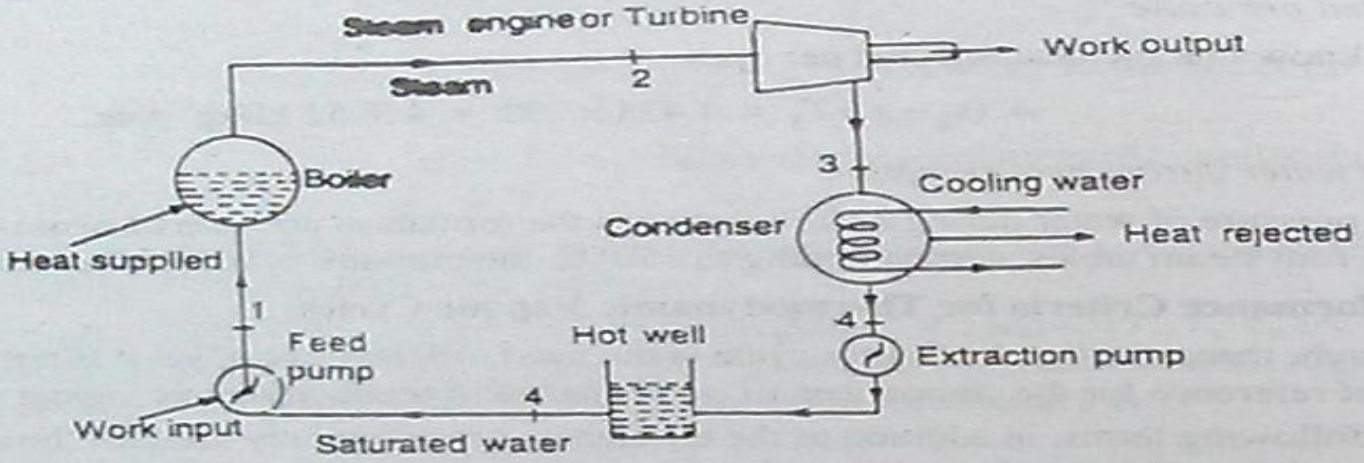
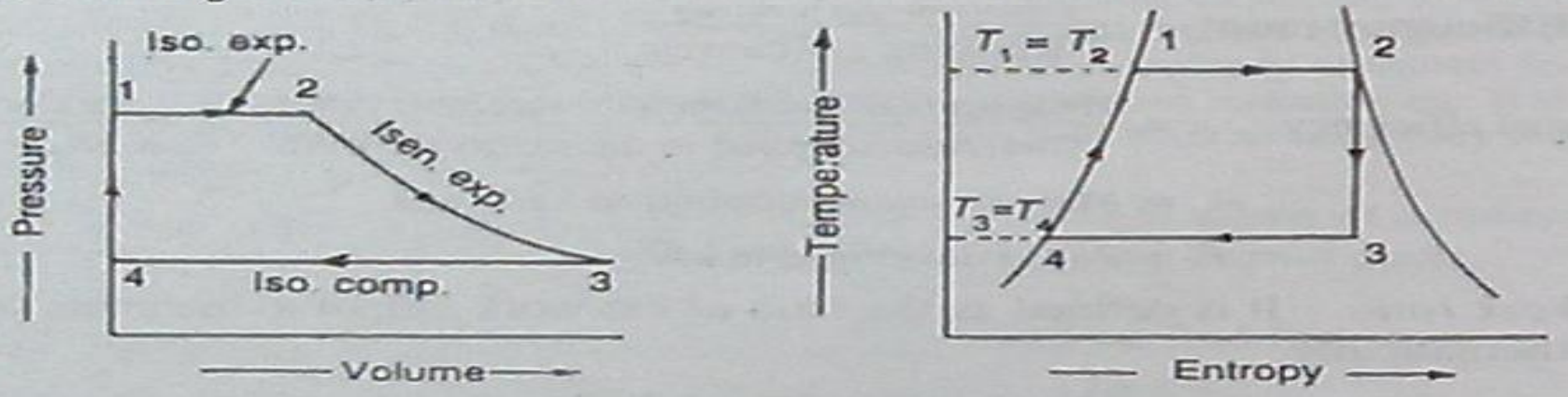


Fig. 10.3. Schematic diagram of a steam engine or turbine plant.

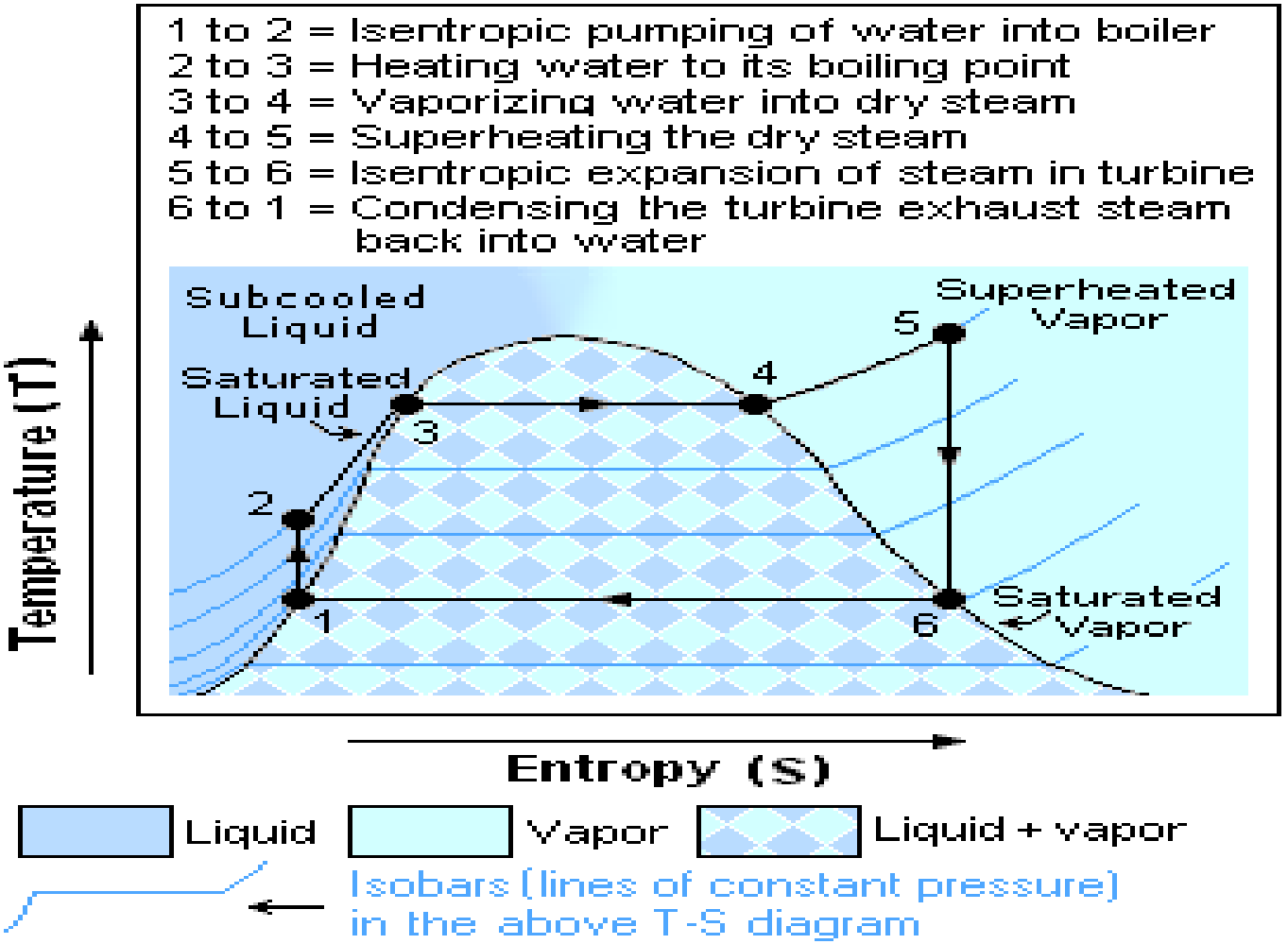
A Carnot cycle, using steam as a working substance, is represented on p-v and T-s diagrams as shown in Fig. 10.4 (a) and (b).



(a) p-v diagram. (b) T-s diagram.

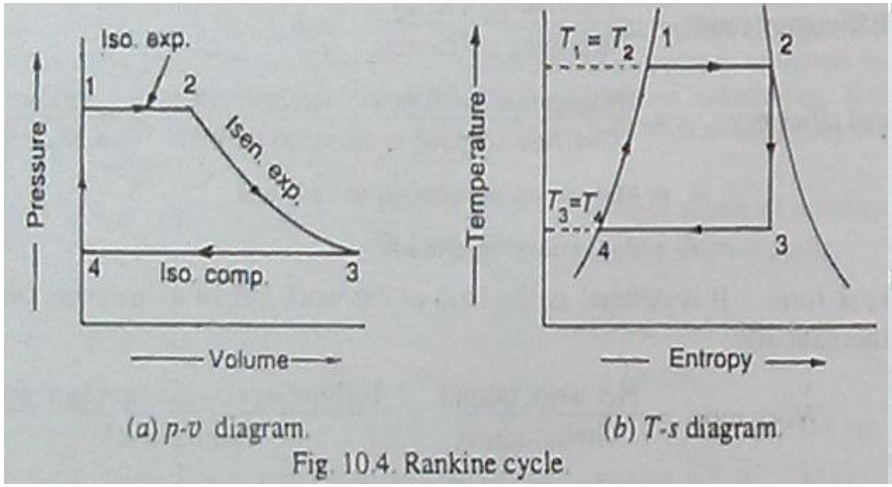
Fig. 10.4. Rankine cycle.

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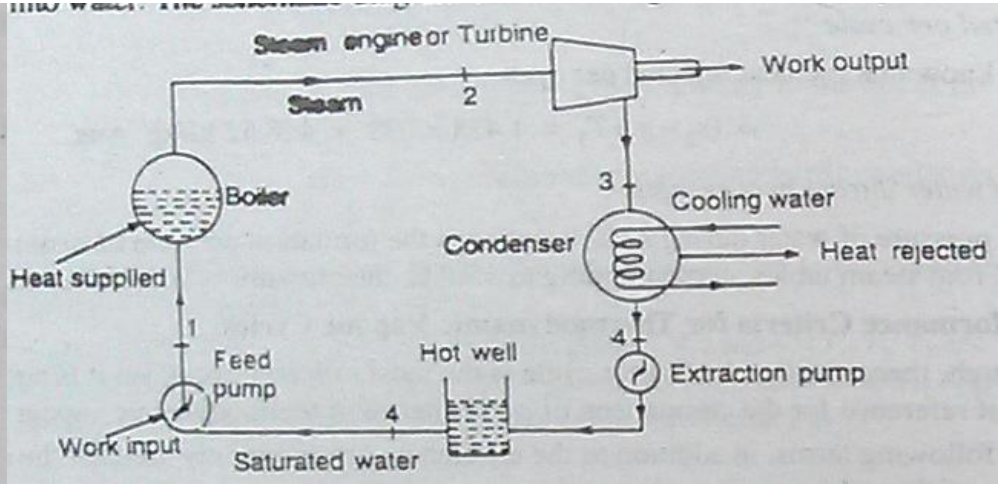
T-S Diagram of Rankine Cycle for Steam to Electric Power

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(a) p-v diagram. (b) T-s diagram.

Fig. 10.4. Rankine cycle.

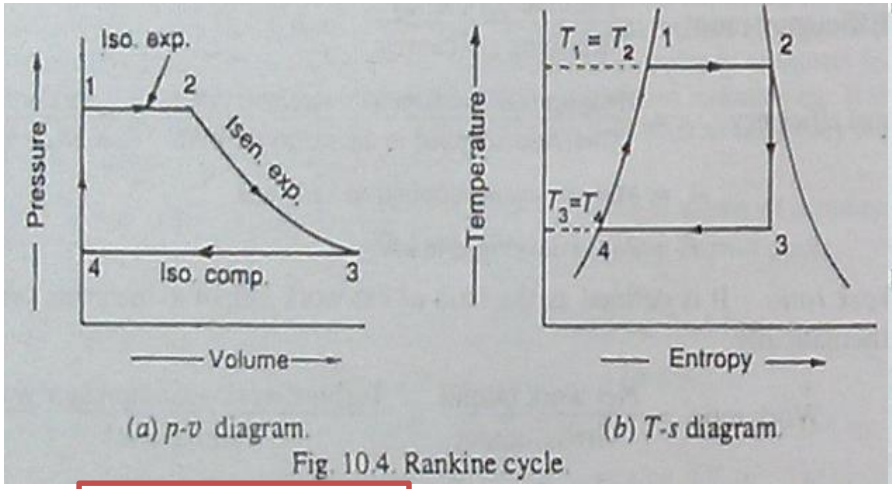


Consider 1 kg of saturated water at pressure p_1 and absolute temperature T_1 as represented by point 1 in Fig. 10.4 (*a*) and (*b*). The cycle is completed by the following four processes :

1. *Process 1-2.* The saturated water at point 1 is isothermally converted into dry saturated steam in a boiler, and the heat is absorbed at a constant temperature T_1 and pressure p_1. The dry state of steam is represented by point 2. It means that the temperature T_2 (*i.e.* at point 2) and pressure p_2 (*i.e.* at point 2) is equal to temperature T_1 and pressure p_1 respectively. This isothermal process is represented by curve 1-2 on p-v and T-s diagrams in Fig. 10.4 (*a*) and (*b*).

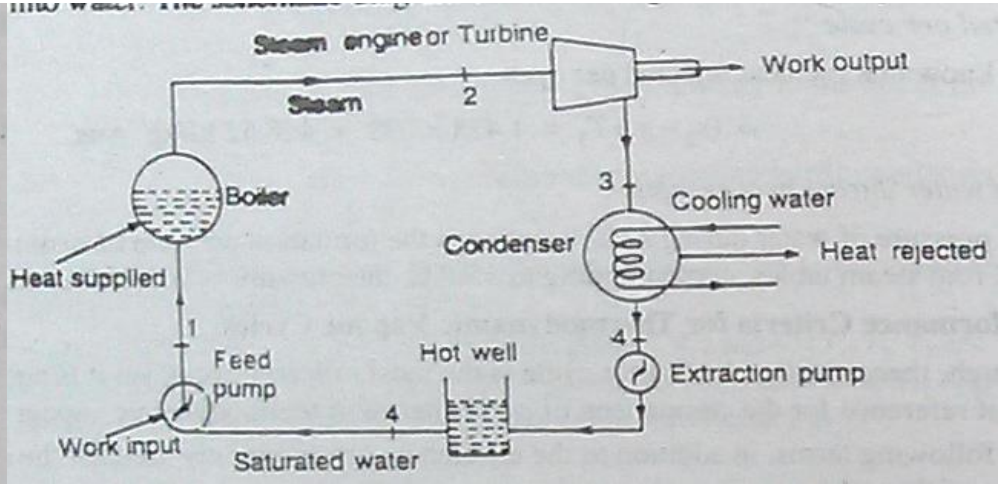
We know that the heat absorbed during isothermal process by water during its conversion into dry steam is its latent heat of vaporisation (*i.e.* $h_{fg1} = h_{fg2}$), corresponding to a pressure p_1 or p_2 ($\because p_1 = p_2$).

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(a) p-v diagram. (b) T-s diagram.

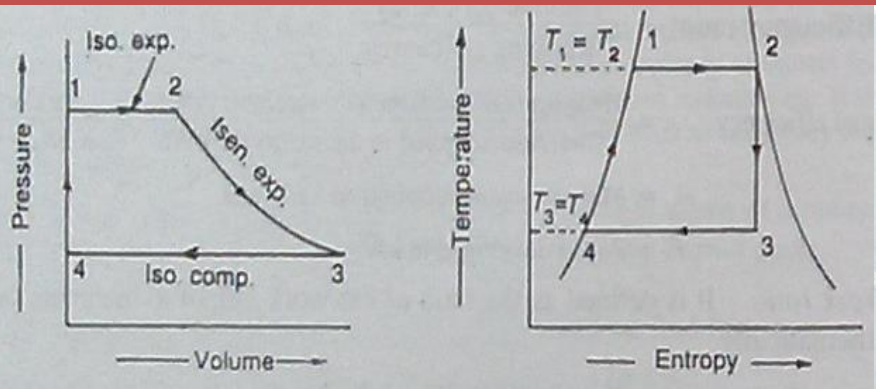
Fig. 10.4. Rankine cycle.



2. *Process 2-3* The dry saturated steam at point 2, now expands isentropically in an engine or turbine. The pressure and temperature falls from p_2 to p_3 and T_2 to T_3 respectively with a dryness fraction x_3. Since no heat is supplied or rejected during this process, therefore there is no change of entropy. The isentropic expansion is represented by the curve 2-3 as shown in Fig. 10.4 (a) and (b).

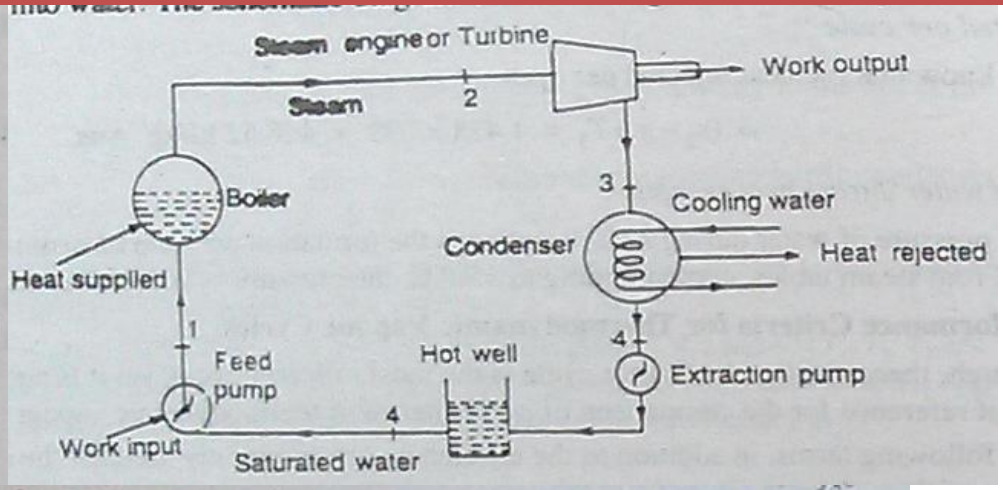
3. *Process 3-4* The wet steam at point 3 is now isothermally condensed in a condenser and the heat is rejected at constant temperature T_3 and pressure p_3 until the whole steam is condensed into water. It means that the temperature T_4 and pressure p_4 (*i.e.* at point 4) is equal to the temperature T_3 and pressure p_3 respectively. The isothermal compression is represented by curve 3-4 on p-v and T-s diagrams in Fig. 10.4 (a) and (b). The heat rejected by steam is its latent heat (equal to $x_3 h_{fg3}$).

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(a) p-v diagram. (b) T-s diagram.

Fig. 10.4. Rankine cycle.



4 Process 4-1 The water at point 4 is now warmed in a boiler at constant volume from temperature T_4 to T_1. Its pressure also rises from p_4 to p_1. This warming operation is represented by the curve 4-1 on p-v and T-s diagrams in Fig. 10.4 (a) and (b). The heat absorbed by water during this operation is equal to the sensible heat or liquid heat corresponding to the pressure p_1 i.e. equal to sensible heat at point 1 *minus* sensible heat at point 4

Let $h_{f1} = h_{f2}$ = Sensible heat or enthalpy of water at point 1 corresponding to a pressure of p_1 or p_2 ($p_1 = p_2$), and

$h_{f4} = h_{f3}$ = Sensible heat or enthalpy of water at point 4 orresponding to a pressure of p_4 or p_3 ($p_4 = p_3$)

Heat absorbed during warming operation 4-1

$$= h_{f1} - h_{f4} = h_{f2} - h_{f3}$$

nd heat absorbed during the complete cycle

= Heat absorbed during isothermal operation 1-2
+ Heat absorbed during warming operation 4-1

$$= h_{fg2} + (h_{f2} - h_{f3}) = h_{f2} + h_{fg2} - h_{f3} = h_2 - h_{f3}$$

(For dry steam, $h_2 = h_{f2} + h_{fg2}$) ... (i)

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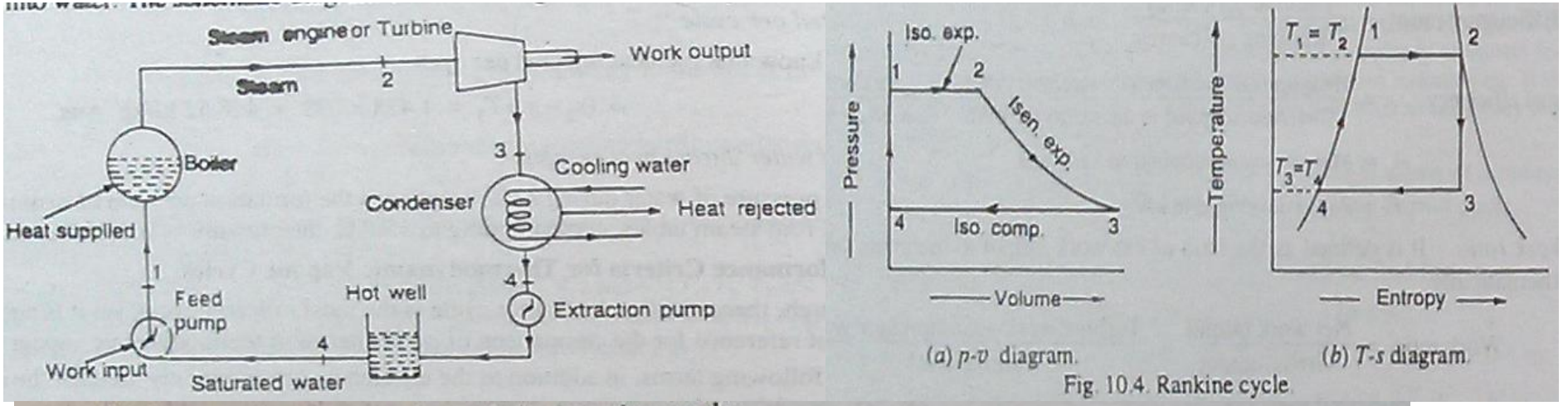


Fig. 10.4. Rankine cycle

We know that heat rejected during the cycle

$$= h_3 - h_{f4} = h_{f3} + x_3 h_{fg3} - h_{f4} = x_3 h_{fg3} \qquad (\because h_3 = h_{f4})$$

∴ *Workdone during the cycle

$$= \text{Heat absorbed} - \text{Heat rejected}$$

$$= (h_2 - h_{f3}) - x_3 h_{fg3}$$

$$= h_2 - (h_{f3} + x_3 h_{fg3}) = h_2 - h_3 \qquad (\because h_3 = h_{f3} + h_{fg3}) \qquad (ii)$$

and efficiency (also called Rankine efficiency),

$$\eta_R = \frac{\text{Work done}}{\text{Heat absorbed}} = \frac{h_2 - h_3}{h_2 - h_{f3}} \qquad (iii)$$

engine are not possible as the various moving parts of the engine, and also the various processes are not reversible. This is obvious, that it is impossible to realise Carnot's engine in actual practice. However, such an imaginary engine is used as the ultimate standard of comparison of all steam engines.

Example 10.1. *A power plant is supplied with dry saturated steam at a pressure of 16 bar and exhausts at 0.2 bar. Using steam tables, find the efficiency of the Carnot cycle.*

Solution. Given : $p_1 = 16$ bar ; $p_2 = 0.2$ bar

From steam tables, corresponding to a pressure of 16 bar, we find that

$$T_1 = 201.4^\circ C = 201.4 + 273 = 474.4\,K$$

and corresponding to a pressure of 0.2 bar, we find that

$$T_3 = 60.1^\circ C = 60.1 + 273 = 333.1\,K$$

We know that efficiency of the Carnot cycle,

$$\eta = \frac{T_1 - T_3}{T_1} = \frac{474.4 - 333.1}{474.4} = 0.298 \quad \text{or} \quad 29.8\% \quad \textbf{Ans.}$$

Example 10.2. *In a Carnot cycle, heat is supplied at 350° C and is rejected at 25° C. The working fluid is water, which while receiving heat, evaporates from liquid at 350° C to steam at 350° C. From the steam tables, the entropy change for this process is 1.438 kJ/kg K.*

If the cycle operates on a stationary mass of 1 kg of water, find the heat supplied, work done and heat rejected per cycle. What is the pressure of water during heat reception ?

Solution. Given : $T_1 = 350^\circ C = 350 + 273 = 623$ K ; $T_3 = 25^\circ C = 25 + 273 = 298$ K ; $s_2 - s_1 = 1.438$ kJ/kg K

Heat supplied per cycle

We know that the heat supplied per cycle,

$$= (s_2 - s_1)\, T_1 = 1.438 \times 623 = 895.87 \text{ kJ/kg} \quad \textbf{Ans.}$$

Work done per cycle

We know that the work done per cycle,

$$= (s_2 - s_1)(T_1 - T_3) = 1.438\,(623 - 298) = 467.35 \text{ kJ/kg}$$ **Ans.**

Heat rejected per cycle

We know that the heat rejected per cycle

$$= (s_2 - s_1)\,T_3 = 1.438 \times 298 = 428.52 \text{ kJ/kg}$$ **Ans.**

Pressure of water during heat reception

The pressure of water during heat reception is the formation pressure of steam corresponding to 350° C. From steam tables, corresponding to 350° C, the pressure is 165.35 bar. Ans.

10.3. Performance Criteria for Thermodynamic Vapour Cycles

Though, theoretically, the Carnot cycle is the most efficient cycle, yet it is not considered as a standard of reference for the comparison of performance of thermodynamic vapour cycles.

The following terms, in addition to the efficiency, are commonly used for the comparison of performance of thermodynamic vapour cycles.

1. *Efficiency ratio.* It is also known as relative efficiency. It is defined as the ratio of thermal efficiency (or actual cycle efficiency) to Rankine efficiency (or ideal cycle efficiency). Mathematically,

$$\text{Efficiency ratio} = \frac{\text{Thermal efficiency}}{\text{* Rankine efficiency}}$$

Note : Thermal efficiency $= \dfrac{\text{Heat equivalent to one kilowatt hour (kWh)}}{\text{Total heat supplied to the steam per kWh}} = \dfrac{3600 \times P}{m_s\,(h_2 - h_3)}$

...lied in kg/h, and

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TABLE A-3 Properties of Saturated Water (Liquid–Vapor): Pressure Table

Press. bar	Temp. °C	Specific Volume m^3/kg Sat. Liquid $v_f \times 10^3$	Specific Volume m^3/kg Sat. Vapor v_g	Internal Energy kJ/kg Sat. Liquid u_f	Internal Energy kJ/kg Sat. Vapor u_g	Enthalpy kJ/kg Sat. Liquid h_f	Enthalpy kJ/kg Evap. h_{fg}	Enthalpy kJ/kg Sat. Vapor h_g	Entropy kJ/kg · K Sat. Liquid s_f	Entropy kJ/kg · K Sat. Vapor s_g	Press. bar
0.04	28.96	1.0040	34.800	121.45	2415.2	121.46	2432.9	2554.4	0.4226	8.4746	0.04
0.06	36.16	1.0064	23.739	151.53	2425.0	151.53	2415.9	2567.4	0.5210	8.3304	0.06
0.08	41.51	1.0084	18.103	173.87	2432.2	173.88	2403.1	2577.0	0.5926	8.2287	0.08
0.10	45.81	1.0102	14.674	191.82	2437.9	191.83	2392.8	2584.7	0.6493	8.1502	0.10
0.20	60.06	1.0172	7.649	251.38	2456.7	251.40	2358.3	2609.7	0.8320	7.9085	0.20
0.30	69.10	1.0223	5.229	289.20	2468.4	289.23	2336.1	2625.3	0.9439	7.7686	0.30
0.40	75.87	1.0265	3.993	317.53	2477.0	317.58	2319.2	2636.8	1.0259	7.6700	0.40
0.50	81.33	1.0300	3.240	340.44	2483.9	340.49	2305.4	2645.9	1.0910	7.5939	0.50
0.60	85.94	1.0331	2.732	359.79	2489.6	359.86	2293.6	2653.5	1.1453	7.5320	0.60
0.70	89.95	1.0360	2.365	376.63	2494.5	376.70	2283.3	2660.0	1.1919	7.4797	0.70
0.80	93.50	1.0380	2.087	391.58	2498.8	391.66	2274.1	2665.8	1.2329	7.4346	0.80
0.90	96.71	1.0410	1.869	405.06	2502.6	405.15	2265.7	2670.9	1.2695	7.3949	0.90
1.00	99.63	1.0432	1.694	417.36	2506.1	417.46	2258.0	2675.5	1.3026	7.3594	1.00
1.50	111.4	1.0528	1.159	466.94	2519.7	467.11	2226.5	2693.6	1.4336	7.2233	1.50
2.00	120.2	1.0605	0.8857	504.49	2529.5	504.70	2201.9	2706.7	1.5301	7.1271	2.00
2.50	127.4	1.0672	0.7187	535.10	2537.2	535.37	2181.5	2716.9	1.6072	7.0527	2.50
3.00	133.6	1.0732	0.6058	561.15	2543.6	561.47	2163.8	2725.3	1.6718	6.9919	3.00
3.50	138.9	1.0786	0.5243	583.95	2546.9	584.33	2148.1	2732.4	1.7275	6.9405	3.50
4.00	143.6	1.0836	0.4625	604.31	2553.6	604.74	2133.8	2738.6	1.7766	6.8959	4.00
4.50	147.9	1.0882	0.4140	622.25	2557.6	623.25	2120.7	2743.9	1.8207	6.8565	4.50
5.00	151.9	1.0926	0.3749	639.68	2561.2	640.23	2108.5	2748.7	1.8607	6.8212	5.00
6.00	158.9	1.1006	0.3157	669.90	2567.4	670.56	2086.3	2756.8	1.9312	6.7600	6.00
7.00	165.0	1.1080	0.2729	696.44	2572.5	697.22	2066.3	2763.5	1.9922	6.7080	7.00
8.00	170.4	1.1148	0.2404	720.22	2576.8	721.11	2048.0	2769.1	2.0462	6.6628	8.00
9.00	175.4	1.1212	0.2150	741.83	2580.5	742.83	2031.1	2773.9	2.0946	6.6226	9.00
10.0	179.9	1.1273	0.1944	761.68	2583.6	762.81	2015.3	2778.1	2.1387	6.5863	10.0
15.0	198.3	1.1539	0.1318	843.16	2594.5	844.84	1947.3	2792.2	2.3150	6.4448	15.0
20.0	212.4	1.1767	0.09963	906.44	2600.3	908.79	1890.7	2799.5	2.4474	6.3409	20.0
25.0	224.0	1.1973	0.07998	959.11	2603.1	962.11	1841.0	2803.1	2.5547	6.2575	25.0
30.0	233.9	1.2165	0.06668	1004.8	2604.1	1008.4	1795.7	2804.2	2.6457	6.1869	30.0
35.0	242.6	1.2347	0.05707	1045.4	2603.7	1049.8	1753.7	2803.4	2.7253	6.1253	35.0
40.0	250.4	1.2522	0.04978	1082.3	2602.3	1087.3	1714.1	2801.4	2.7964	6.0701	40.0
45.0	257.5	1.2692	0.04406	1116.2	2600.1	1121.9	1676.4	2798.3	2.8610	6.0199	45.0
50.0	264.0	1.2859	0.03944	1147.8	2597.1	1154.2	1640.1	2794.3	2.9202	5.9734	50.0
60.0	275.6	1.3187	0.03244	1205.4	2589.7	1213.4	1571.0	2784.3	3.0267	5.8892	60.0
70.0	285.9	1.3513	0.02737	1257.6	2580.5	1267.0	1505.1	2772.1	3.1211	5.8133	70.0
80.0	295.1	1.3842	0.02352	1305.6	2569.8	1316.6	1441.3	2758.0	3.2068	5.7432	80.0
90.0	303.4	1.4178	0.02048	1350.5	2557.8	1363.3	1378.9	2742.1	3.2858	5.6772	90.0
100.	311.1	1.4524	0.01803	1393.0	2544.4	1407.6	1317.1	2724.7	3.3596	5.6141	100.
110.	318.2	1.4886	0.01599	1433.7	2529.8	1450.1	1255.5	2705.6	3.4295	5.5527	110.

BME lecture week:3

Example 10.3. *A steam power plant is supplied with dry saturated steam at a pressure of 12 bar and exhausts into a condenser at 0.1 bar. Calculate the Rankine efficiency by using 1. Steam tables, and 2. Mollier chart.*

Solution. Given : $x_2 = 1$; $p_2 = 12$ bar , $p_3 = 0.1$ bar

1 *Using steam tables*

From steam tables, corresponding to a pressure of 12 bar, we find that

$$h_{f2} = 798.4 \text{ kJ/kg} , \; h_{fg2} = 1984.3 \text{ kJ/kg} , \; s_2 = s_{g2} = 6.519 \text{ kJ/kg K}$$

and corresponding to a pressure of 0.1 bar, we find that

$$h_{f3} = 191.8 \text{ kJ/kg} ; \; h_{fg3} = 2393 \text{ kJ/kg} , \; s_{f3} = 0.649 \text{ kJ/kg K} ; \; s_{fg3} = 7.502 \text{ kJ/kg K}$$

First of all, let us find the dryness fraction (x_3) of steam after isentropic expansion. We know that in isentropic expansion,

Entropy before expansion (s_2) = Entropy after expansion (s_3)

$$6.519 = s_{f3} + x_3 \, s_{fg3} = 0.649 + x_3 \times 7.502$$

$$\therefore \quad x_3 = 0.783$$

Now enthalpy or total heat of dry saturated steam before expansion,

$$h_2 = h_{g2} = h_{f2} + x_2 \, h_{fg2} = 789.4 + 1 \times 1984.3 = 2782.7 \text{ kJ/kg}$$

and enthalpy or total heat of wet steam after expansion,

$$h_3 = h_{f3} + x_3 \, h_{fg3} = 191.8 + 0.783 \times 2393 = 2065.5 \text{ kJ/kg}$$

We know that Rankine efficiency,

$$\eta_R = \frac{h_2 - h_3}{h_2 - h_{f3}} = \frac{2782.7 - 2065.5}{2782.7 - 191.8} = 0.277 \quad \text{or} \quad 27.7 \% \quad \textbf{Ans.}$$

BME lecture week:3

Example 10.3. *A steam power plant is supplied with dry saturated steam at a pressure of 12 bar and exhausts into a condenser at 0.1 bar. Calculate the Rankine efficiency by using 1. Steam tables, and 2. Mollier chart.*

Solution. Given : $x_2 = 1$; $p_2 = 12$ bar , $p_3 = 0.1$ bar

2. Using Mollier chart

In Fig. 10.5, the initial condition of steam is marked at point 2 and final condition at point 3. From Mollier chart, we find that

$h_2 = 2780$ kJ/kg , $h_3 = 2060$ kJ/kg , and

$h_{f3} = 191.8$ kJ/kg (From steam tables)

We know that the Rankine efficiency,

$$\eta_R = \frac{h_2 - h_3}{h_2 - h_{f3}} = \frac{2780 - 2060}{2780 - 191.8}$$

$= 0.278$ or 27.8% **Ans.**

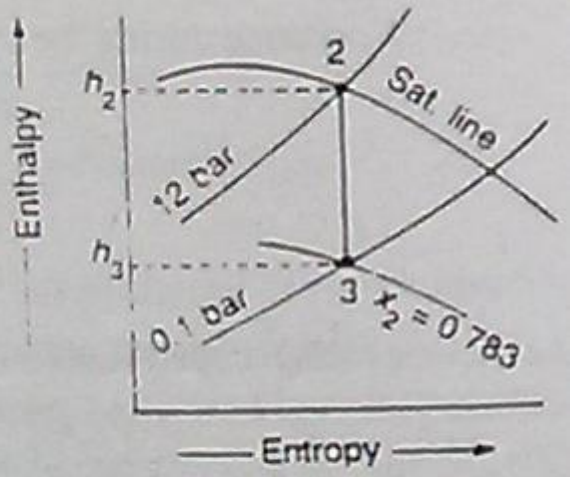
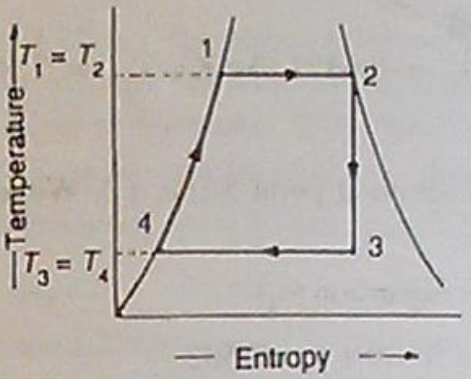


Fig. 10.5



(*a*) *T-s* diagram

BME lecture week:3

Please study
-Chapter (7, 8) and (13, 14, 15) for week one and two,
-Chapter 10 for week three,
-Chapter 6 and 26 for week four,
-Chapter 25 for week five,
These are the syllabus for mid term.

Thank you for your attention